Lecture 12

Integrability, and what a mechanical system is

Objectives. *Lecture 11 ended with the rigid body moving along the curves where a sphere meets an energy ellipsoid. The pendulum and the central force were solved the same way, by conserved quantities (Lectures 1 and 10). What do n conserved quantities force in n degrees of freedom? One condition beyond conservation is needed,* involution*, a condition on their brackets. A level set on which the n differentials are independent is then an n-dimensional* Lagrangian *surface, invariant under all n flows. Each compact connected component is a torus carrying action–angle variables. The course closes by comparing the three descriptions of a mechanical system and explaining which structures quantum mechanics retains.*

By the end of this lecture you will be able to:

- Test independence and involution of conserved quantities.
- Explain why regular common level sets of a Liouville integrable system are Lagrangian.
- State the Liouville–Arnold theorem for compact connected regular components.
- Compare the Lagrangian, Hamiltonian and Poisson descriptions of a classical system.

12.1 Lagrangian subspaces

Lecture 6 put a bilinear form behind the bracket: $\omega(u,v) = u^{\mathsf{T}}\mathbb{J}v$ on $\mathbb{R}^{2n}$ (Definition 6.11; the matrix has been written $\mathbb{J}$ since Lecture 10). This lecture needs one fact about it: how large a subspace can be on which ω vanishes. For an inner product only the zero subspace qualifies. For ω every line does, since antisymmetry gives $\omega(w,w) = 0$.

Definition 12.1 (ω-orthogonal complement)**.** For a subspace $W \subseteq \mathbb{R}^{2n}$,

$$W^{\omega} = \{v \in \mathbb{R}^{2n} : \omega(v,w) = 0 \text{ for all } w \in W\}$$

is its *ω-orthogonal complement*.

Unlike an orthogonal complement, W^{ω} can contain W, as it does for every line. Dimensions still add up.

Lemma 12.2 (Dimension of the complement)**.** *The equality* $\dim W + \dim W^{\omega} = 2n$ *holds for every subspace* $W \subseteq \mathbb{R}^{2n}$.

Proof. We have $\omega(v,w) = v^{\mathsf{T}}\mathbb{J}w = v \cdot (\mathbb{J}w)$, so W^ω is the orthogonal complement, for the dot product, of $\mathbb{J}W = \{\mathbb{J}w : w \in W\}$. Since $\mathbb{J}$ is invertible, $\dim \mathbb{J}W = \dim W$, and an orthogonal complement in $\mathbb{R}^{2n}$ has dimension $2n - \dim \mathbb{J}W$. ∎

Definition 12.3 (Isotropic and Lagrangian)**.** A subspace $W \subseteq \mathbb{R}^{2n}$ is *isotropic* if ω vanishes on W, that is $W \subseteq W^\omega$, and *Lagrangian* if $W = W^\omega$.

The adjective is named after Lagrange. It has nothing to do with the function $\mathcal{L}$.

If W is isotropic, then $\dim W \le \dim W^\omega = 2n - \dim W$ by Lemma 12.2, so $\dim W \le n$; and an isotropic W is Lagrangian exactly when $\dim W = n$, since then W and W^ω have the same dimension.

For example, for any $n \times n$ matrix B the graph

$$W_B = \{(y, By) : y \in \mathbb{R}^n\} \qquad \text{has} \qquad \omega\big((y,By),(z,Bz)\big) = y \cdot Bz - By \cdot z = y^{\mathsf{T}}(B - B^{\mathsf{T}})z$$

and dimension n, so W_B is Lagrangian exactly when B is symmetric. The q-plane $\{p = 0\}$ is $B = 0$; the p-plane $\{q = 0\}$ is Lagrangian too.

12.2 Integrals in involution

Throughout the rest of the lecture, $\mathcal{M} \subseteq \mathbb{R}^{2n}$ is open, with the canonical bracket of Definition 5.2, and $H \in C^\infty(\mathcal{M})$.

Proposition 5.11 says that F is conserved exactly when $\{F, H\} = 0$, and a conserved F confines each motion to a level set of F. Collect several conserved quantities into one map

$$F = (F_1, \dots, F_k) \colon \mathcal{M} \to \mathbb{R}^k, \qquad \mathcal{M}_c = F^{-1}(c) = \{x \in \mathcal{M} : F_i(x) = c_i \text{ for all } i\}, \tag{12.1}$$

and every motion stays in the level set $\mathcal{M}_c$ through its initial point. Two conditions refine this picture, and each has a name.

Definition 12.4 (Independence)**.** The functions $F_1, \dots, F_k$ are *independent at* x if $\mathrm{d}F_1(x), \dots, \mathrm{d}F_k(x)$ are linearly independent — equivalently, if the $k \times 2n$ matrix $DF(x)$ has rank k. They are *independent* if this holds at every point of an open dense subset of $\mathcal{M}$.

A subset of $\mathcal{M}$ is *dense* if every small open ball in $\mathcal{M}$ contains a point of it. The definition asks for an open dense set, not for all of $\mathcal{M}$, because the examples need it: for the oscillators of Example 12.8 independence fails exactly where one of them is at rest.

Definition 12.5 (Involution)**.** The functions $F_1, \dots, F_k$ are *in involution* if $\{F_i, F_j\} = 0$ for all i, j.

Involution is a genuine restriction and not a consequence of conservation. For a central force the three components of $\mathbf{L} = q \times p$ are conserved, and by (5.11) $\{L_1, L_2\} = L_3$, which is not identically zero.

A different choice of three works. Take the constants H, $|\mathbf{L}|^2$ and L_3 of Proposition 9.4. Both H and $|\mathbf{L}|^2$ are scalars (Proposition 5.14), so they have zero bracket with L_3. Also $|\mathbf{L}|^2$ is conserved, so $\{|\mathbf{L}|^2, H\} = 0$. These three are in involution. In spherical coordinates they are (9.12), $p_\vartheta^2 + p_\varphi^2/\sin^2\vartheta$ and p_φ. Their derivative in $(p_r, p_\vartheta, p_\varphi)$ is a triangular matrix with diagonal entries p_r/m, $2p_\vartheta$ and 1. So they are independent wherever $p_r \neq 0$ and $p_\vartheta \neq 0$, and that set is open and dense.

How many independent functions can be in involution at once? The dimension bound for isotropic subspaces gives the answer. It explains why Liouville integrability requires exactly n functions.

Proposition 12.6 (At most n in involution)**.** *Let $F_1, \dots, F_k \in C^\infty(\mathcal{M})$ be in involution and independent at a point x. Then $k \le n$.*

Proof. The vectors $X_{F_i}(x) = \mathbb{J}\nabla F_i(x)$ are linearly independent, because $\mathbb{J}$ is invertible and the $\nabla F_i(x)$ are. Let $W = \operatorname{span}\{X_{F_1}(x), \dots, X_{F_k}(x)\}$, of dimension k. By (6.9),

$$\omega\big(X_{F_i}(x), X_{F_j}(x)\big) = \big\{F_i, F_j\big\}(x) = 0,$$

so by bilinearity ω vanishes on W: W is isotropic, and $k = \dim W \le n$ by Definition 12.3 and the remark after it. ∎

Definition 12.7 (Liouville integrable)**.** The system $(\mathcal{M}, H)$ is *Liouville integrable* if there are $F_1 = H, F_2, \dots, F_n \in C^\infty(\mathcal{M})$ that are in involution and independent in the sense of Definition 12.4.

Each F_i is then conserved, since $\{F_i, H\} = \{F_i, F_1\} = 0$. By Proposition 12.6 no family of more than n functions in involution is independent at any point, so n is the largest number available.

Example 12.8 (n uncoupled oscillators)**.** On $\mathbb{R}^{2n}$ take $F_k = \frac{1}{2}\big(p_k^2 + \nu_k^2 q_k^2\big)$ with $\nu_k > 0$, and $H = \sum_k F_k$. Distinct F_k involve disjoint variables, so $\big\{F_i, F_j\big\} = 0$. Also $\mathrm{d}F_k = \nu_k^2 q_k\,\mathrm{d}q_k + p_k\,\mathrm{d}p_k$, which involves the k-th pair of coordinates and no other: the k-th row of DF is $(\nu_k^2 q_k, p_k)$ in the columns of that pair and zero elsewhere, and it vanishes exactly when $q_k = p_k = 0$. Rows with disjoint supports are linearly independent unless one of them is zero, so $\operatorname{rank} DF = n$ off the union of the n planes $P_k = \{q_k = p_k = 0\}$. Each P_k is a linear subspace of dimension $2n-2$, so the complement of their union is open and dense. So the F_k are independent. The family $H, F_2, \dots, F_n$ of Definition 12.7 is too: its brackets vanish by bilinearity, and since $\mathrm{d}H = \sum_k \mathrm{d}F_k$ the two lists of differentials span the same space at every point. So the system is integrable.

12.3 Level sets are Lagrangian

The tangent spaces of a regular level set are determined by the identities from Lecture 6, (6.9): $\omega(X_f, v) = \nabla f \cdot v$ and $\omega(X_f, X_g) = \{f, g\}$.

Theorem 12.9 (Regular level sets are Lagrangian)**.** *Let $F = (F_1, \dots, F_n) \colon \mathcal{M} \to \mathbb{R}^n$ be C^∞ with $\big\{F_i, F_j\big\} = 0$ on $\mathcal{M}$, let $c \in \mathbb{R}^n$, and suppose $\mathrm{d}F_1, \dots, \mathrm{d}F_n$ are linearly independent at every point of $\mathcal{M}_c = F^{-1}(c)$. Then $\mathcal{M}_c$ is a surface of dimension n (Definition 2.3) or empty, and at every $x \in \mathcal{M}_c$*

$$T_x\mathcal{M}_c = \ker DF(x) = \operatorname{span}\big\{X_{F_1}(x), \dots, X_{F_n}(x)\big\}, \tag{12.2}$$

and ω vanishes on $T_x\mathcal{M}_c$, which is therefore a Lagrangian subspace in the sense of Definition 12.3. Moreover every X_{F_i} is tangent to $\mathcal{M}_c$, and $\mathcal{M}_c$ is invariant under each of their flows — in particular under the flow of $H = F_1$.

Proof. The hypothesis is that c is a regular value of F (Definition 2.1), so $\mathcal{M}_c$ is a surface of dimension $2n - n = n$ by Corollary 2.4, with $T_x\mathcal{M}_c = \ker DF(x)$ by Theorem 2.7.

Each $X_{F_i}(x)$ lies in that kernel: by (6.9),

$$\nabla F_j(x) \cdot X_{F_i}(x) = \omega\big(X_{F_j}(x), X_{F_i}(x)\big) = \big\{F_j, F_i\big\}(x) = 0 \qquad \text{for every } j.$$

The n vectors $X_{F_i}(x) = \mathbb{J}\nabla F_i(x)$ are linearly independent, $\mathbb{J}$ being invertible, and $\ker DF(x)$ has dimension n; so they are a basis of it, which is (12.2).

On that basis $\omega(X_{F_i}, X_{F_j}) = \big\{F_i, F_j\big\} = 0$, so by bilinearity ω vanishes on $T_x\mathcal{M}_c$; a subspace of dimension n on which ω vanishes is Lagrangian.

For invariance, let γ be a motion of X_{F_i} with $\gamma(0) \in \mathcal{M}_c$. Then $\frac{\mathrm{d}}{\mathrm{d}t} F_j(\gamma(t)) = \big\{F_j, F_i\big\}(\gamma(t)) = 0$ for every j, so $F(\gamma(t)) = c$ for as long as γ exists. ∎

The involution hypothesis is essential. Take $F_1 = q_1$ and $F_2 = p_1$ on $\mathbb{R}^4$. The $\mathrm{d}F_i$ are independent everywhere, every level set is a plane of dimension $n = 2$, and every hypothesis of Theorem 12.9 holds except $\{F_1, F_2\} = 0$. The level set $\{q_1 = a,\ p_1 = b\}$ is the (q_2, p_2) plane, which contains e_{q_2} and e_{p_2} with $\omega(e_{q_2}, e_{p_2}) = 1$: it is not Lagrangian. Nor is it invariant under the flow of F_1, which is $p_1 \mapsto p_1 - t$.

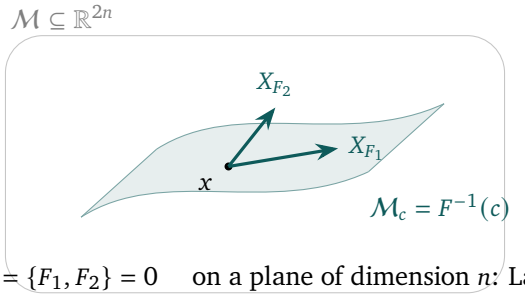


Figure 12.1: Theorem 12.9 in two degrees of freedom. The n fields X_{F_i} are tangent to the level set $\mathcal{M}_c$ and span its tangent plane; ω evaluated on any two of them is the bracket of the corresponding integrals, which is zero, so the tangent plane is Lagrangian.

12.4 The Liouville–Arnold theorem

Theorem 12.9 says what a regular level set looks like near each of its points. What it is as a whole is decided by the flows. By Theorem 6.7, $\big[X_{F_i}, X_{F_j}\big] = -X_{\{F_i,F_j\}} = 0$: the n fields tangent to $\mathcal{M}_c$ commute. The proof of the next theorem starts from there, and ends by recognising a compact connected component as a torus; that proof is the part this course cites.

Theorem 12.10 (Liouville–Arnold · *cited*)**.** *Let $\mathcal{M} \subseteq \mathbb{R}^{2n}$ be open and let $F_1 = H, F_2, \dots, F_n$ be C^∞ and in involution. Fix $c \in \mathbb{R}^n$, let Σ be a connected component of $\mathcal{M}_c$, and suppose*

(I) *$\mathrm{d}F_1, \dots, \mathrm{d}F_n$ are linearly independent at every point of Σ, and*

(C) *Σ is compact.*

Then

1. *Σ is an n-dimensional Lagrangian surface, invariant under the flow of every X_{F_i};*
2. *Σ is diffeomorphic to the torus $T^n = \mathbb{R}^n/2\pi\mathbb{Z}^n$, that is, to n angles $\varphi_1, \dots, \varphi_n$, each defined modulo 2π;*
3. *in those coordinates the motion of H is $\varphi(t) = \varphi(0) + \nu t$, with a frequency vector*

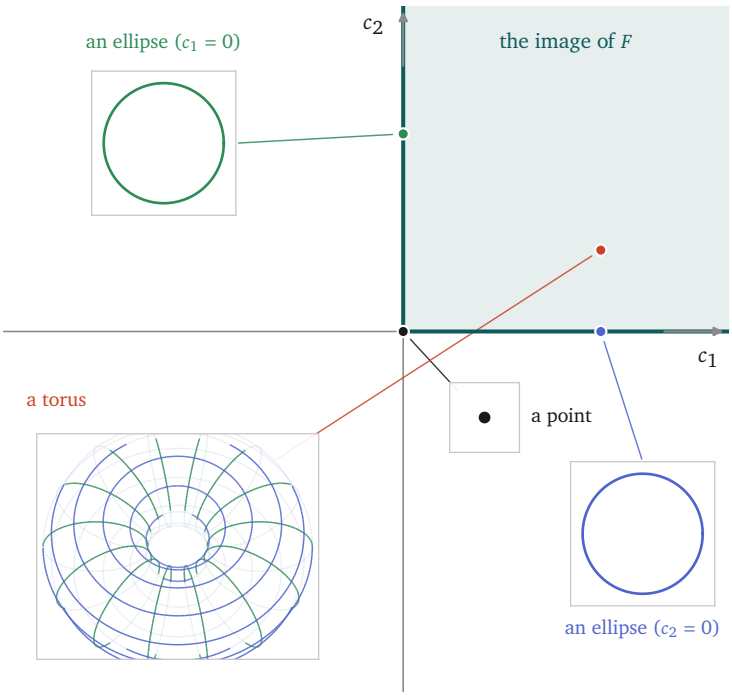


Figure 12.2: The integral map $F = (F_1, F_2)$ of two oscillators: its image, shaded, and the level set over each marked value. The torus is drawn as a torus of revolution, a picture of the product of the two ellipses and not a projection from $\mathbb{R}^4$. Its blue circles are copies of the first ellipse, which carry I_1, and its green ones copies of the second, which carry I_2.

> $\nu \in \mathbb{R}^n$ *depending only on* c*;*
>
> 4. *there is a neighbourhood of* Σ *carrying canonical coordinates* (φ, I) — action–angle variables — *in which the* I_k *are functions of* c *alone,* H *is a function of* I *alone and*
>
> $$I_k = \frac{1}{2\pi} \oint_{\gamma_k} p \cdot \mathrm{d}q, \qquad \dot{\varphi}_k = \nu_k(I) = \frac{\partial H}{\partial I_k}, \tag{12.3}$$
>
> *for suitable oriented closed curves* $\gamma_1, \dots, \gamma_n$ *on* Σ.

Part 1 is Theorem 12.9, proved above, applied on the open set where the $\mathrm{d}F_i$ are independent, which contains Σ. The rest is Arnold's §§49–50 (pp. 271–285), where the proof is given and which the *Integrability* course develops; Arnold states the theorem on symplectic manifolds, and (12.3) is how he writes the actions (§50 C, p. 282).

Remark 12.11 (Closedness of the canonical form on a level set)**.** The action integral must be unchanged when its closed curve is continuously deformed inside Σ. We prove one ingredient of this invariance here. Let $r : V \to \mathbb{R}^{2n}$, $V \subseteq \mathbb{R}^n$, be a local parametrisation of Σ (Definition 2.3), and write $r(s) = (q(s), p(s))$. Substituting it turns $p \cdot \mathrm{d}q$ into $\sum_j a_j \, \mathrm{d}s_j$ with $a_j = p \cdot \partial_j q$, and

$$\partial_i a_j - \partial_j a_i = \partial_i p \cdot \partial_j q - \partial_j p \cdot \partial_i q = -\omega\big(\partial_i r, \partial_j r\big) = 0,$$

because $\partial_i r$ and $\partial_j r$ are tangent to Σ, on which ω vanishes (Theorem 12.9). So on Σ the form $p \cdot \mathrm{d}q$ satisfies the closedness condition (8.6). That the loop integral is then unchanged when the loop is deformed *inside* Σ needs Stokes' theorem on Σ (Arnold, p. 283, Figure 218),

which this course does not build, and it is part of the citation.

Example 12.12 (Action–angle for the oscillators)**.** For Example 12.8 the level set $\{F_1 = c_1, \dots, F_n = c_n\}$ with every $c_k > 0$ is the product of the n ellipses $p_k^2 + \nu_k^2 q_k^2 = 2c_k$, one in each plane: a torus, explicitly and with no theorem needed. (The level sets of the family $H, F_2, \dots, F_n$ are the same sets with other labels.) The natural closed curve γ_k runs once round the k-th ellipse, the other pairs held fixed, in the direction in which the k-th oscillator moves, and

$$I_k = \frac{1}{2\pi} \oint_{\gamma_k} p_k \, \mathrm{d}q_k = \frac{\text{area enclosed}}{2\pi} = \frac{c_k}{\nu_k},$$

which is (9.18), one degree of freedom at a time. Then $H = \sum_k c_k = \sum_k \nu_k I_k$ and $\partial H / \partial I_k = \nu_k$: the frequencies are the ν_k on every torus. That is special to the oscillators, whose motion is linear, so that the frequencies cannot depend on the amplitude.

For $n = 2$ the whole family fits in one picture (Figure 12.2). The values of $F = (F_1, F_2)$ fill the closed quadrant $c_1 \geq 0$, $c_2 \geq 0$, and over an interior point the level set is a torus. Over a point of an edge, say $c_2 = 0 < c_1$, every point of the level set lies in the plane P_2, where $\mathrm{d}F_2 = 0$: hypothesis (I) fails, and with it the independence that Theorem 12.9 asks for. The conclusion fails too. The level set is the single ellipse $p_1^2 + \nu_1^2 q_1^2 = 2c_1$ in the plane $q_2 = p_2 = 0$, a curve and not a surface of dimension 2; over the corner it is the origin. Hypothesis (C) is needed as well. For the free particle, $n = 1$ and $H = \frac{1}{2}p^2$, every value $c > 0$ is regular, and the level set is the pair of lines $p = \pm\sqrt{2c}$: each component is a line, not a circle.

12.5 What a mechanical system is: the final table

Lecture 1 asked what a classical mechanical system is and split the question into three: what is a state, what is an observable, how do they evolve? It gave provisional answers for Newtonian systems on $\mathbb{R}^{2n}$. Lecture 3 gave the first answer: a configuration space with a Lagrangian. Lecture 7 gave the second, Definition 7.11: a Poisson algebra of observables, a distinguished H, and a convex set of states. The third answer compares the three layers. Each layer passes to the next where the passage is possible, and each later layer covers more systems. The table puts the three side by side, and every entry of it has now been proved or cited.

	Lagrange	**Hamilton**	**Poisson**
Arena	Q, a surface (Definition 2.3)	$\mathcal{M}$ with a canonical atlas (Definition 11.10)	a Poisson algebra $\mathcal{A}$ (Definition 5.7)
A state	$(q, \dot{q})$, and the curve is the object	a point of $\mathcal{M}$ (Corollary 1.7: one orbit through each)	a positive normalised expectation, on the observables it can read; for the states of Definition 7.1, pure = Dirac = dispersion-free = extreme (Proposition 7.3)
The law	$\delta \int \mathcal{L} \, \mathrm{d}t = 0$ with fixed ends (Theorem 3.4)	$\dot{x} = \mathbb{J}\nabla H$ (Theorem 4.5)	$\dot{f} = \{f, H\}$ (Theorem 5.15); for a smooth density in canonical coordinates, $\partial_t \rho = \{H, \rho\}$ (Theorem 7.6)

	Lagrange	**Hamilton**	**Poisson**
Symmetry	point transformations; Noether (Theorem 3.14)	canonical maps (Theorem 8.2); momentum maps (Definition 10.5)	automorphisms of $\{\cdot,\cdot\}$; for a bracket on an open subset of $\mathbb{R}^m$, Jacobi $\Leftrightarrow$ the complete flows preserve it (Theorem 6.5)
Conserved	from invariance of $\mathcal{L}$	from a group action with H invariant, J (Theorem 10.9)	$\{F,H\} = 0$ (Proposition 5.11); Casimirs for *every* H (Definition 11.6)
Integrable	—	regular common level sets of n integrals in involution are Lagrangian (Theorem 12.9); their compact connected components are tori (Theorem 12.10)	for the canonical bracket on an open subset of $\mathbb{R}^{2n}$: $H = F_1, \dots, F_n$ independent and in involution; at most n can be independent at a point (Definition 12.7 and Proposition 12.6)
What fails	the Legendre map need not invert (Figure 4.3)	no single canonical chart on the sphere (Proposition 11.9); the rigid body's bracket is degenerate, with leaves the spheres $\lvert\mathbf{M}\rvert = R$ and the origin (Corollary 11.14)	the point: in quantum mechanics no point remains to evaluate at (§ 12.6)

Read across, the description of a state changes, but the classical layers keep the same point states. The Poisson layer represents them algebraically, by Dirac expectations. Quantum mechanics keeps states and loses their identification with points, which is the last entry of the table.

12.6 What quantum mechanics keeps

The algebraic description extends to quantum mechanics. We compare the structures that remain, the correspondence that is exact, and its limits.

What is kept

The right-hand column previews quantum mechanics. Here Q_i and P_j are the position and momentum operators. On functions of x in one degree of freedom, Q is multiplication by x and $P = -\mathrm{i}\hbar\,\mathrm{d}/\mathrm{d}x$, the *Schrödinger operators*. Further, $A^\dagger$ denotes the adjoint of an operator, U is unitary, and $\hbar$ is the reduced Planck constant. A *density operator* $\hat{\rho}$ is the counterpart of a density: a positive operator of trace 1, with expectations $\operatorname{tr}(\hat{\rho}A)$. As in Proposition 7.3, a state is *dispersion-free* if every position and momentum has variance zero in it. The quantum constructions will be developed in the Quantum Mechanics course.

Dictionary

commutative Poisson algebra $C^\infty(\mathcal{M})$ $\longrightarrow$ a non-commutative algebra of operators closed under adjoints, with the Lie bracket $\frac{1}{\mathrm{i}\hbar}\,[A,B]$

$\{q_i, p_j\} = \delta_{ij}$ (5.13) $\longrightarrow$ $\left[Q_i, P_j\right] = \mathrm{i}\hbar\,\delta_{ij}$

$\dot{f} = \{f, H\}$ (Theorem 5.15) $\longrightarrow$ $\dot{A} = \frac{1}{\mathrm{i}\hbar}\,[A,H]$, the Heisenberg equation

$\partial_t \rho = \{H, \rho\}$ for smooth ρ (Theorem 7.6) $\longrightarrow$ $\partial_t \hat{\rho} = \frac{1}{\mathrm{i}\hbar}\,[H, \hat{\rho}]$, von Neumann's equation

the flow acts by Poisson automorphisms (Theorems 6.4 and 6.5) $\longrightarrow$ $A \mapsto U^\dagger A U$ preserves products and adjoints

states are mixtures of densities and points; pure = Dirac = dispersion-free = extreme (Definition 7.1 and Proposition 7.3) $\longrightarrow$ states are density operators; pure = rank one = extreme, but *no* dispersion-free states

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The actions survive too, but only approximately. In one degree of freedom the rule of Bohr and Sommerfeld keeps the orbits with $I = \hbar(m + \frac{1}{2})$, $m = 0, 1, 2, \dots$, for a libration between two simple turning points; for the oscillator, where $E = \nu I$ by (9.18), it gives exactly the quantum energies $\hbar\nu(m + \frac{1}{2})$.

The automorphism entry explains the role of Jacobi in the comparison. By antisymmetry the Jacobi identity can be rewritten as the second form of (5.7):

$$\{\{f, g\}, H\} = \{\{f, H\}, g\} + \{f, \{g, H\}\} \qquad \text{for all } f, g, H: \tag{12.4}$$

each map $f \mapsto \{f, H\}$ is a *derivation* of the bracket, which is what "the flow of H preserves the bracket" says at $t = 0$, and Theorem 6.5 proved that Jacobi holds exactly when every such flow, when complete, preserves it. The rewriting uses bilinearity and antisymmetry and nothing else — not that observables are functions, and not that they commute. So it tells any successor theory what it must supply if every observable, the Hamiltonian among them, is to generate a dynamics by automorphisms: a bracket satisfying Jacobi. It does not say *which* bracket. That the quantum bracket is $\frac{1}{i\hbar}[\cdot, \cdot]$ is quantum mechanics' own input.

Point evaluation has no quantum counterpart. A *character* of an algebra is a linear map χ to the numbers with $\chi(fg) = \chi(f)\chi(g)$ and $\chi(1) = 1$. Evaluation at a point is one, since $(fg)(z) = f(z)g(z)$; so the pure states of Definition 7.1, which are the Dirac states (Proposition 7.3), are characters. The quantum algebra, which contains Q and P with $[Q, P] = i\hbar\,\mathbb{1}$, has none. A character would give $\chi([Q, P]) = \chi(Q)\chi(P) - \chi(P)\chi(Q) = 0$, but also $\chi([Q, P]) = \chi(i\hbar\,\mathbb{1}) = i\hbar \neq 0$. The state survives; the point does not.

What is matched exactly

An *ideal* of a Lie algebra $\mathfrak{g}$ is a subspace $\mathfrak{i}$ whose bracket with every element of $\mathfrak{g}$ lies in $\mathfrak{i}$. Suppose that $\mathfrak{g} = \mathfrak{i} \oplus \mathfrak{s}$ as a vector space, with $\mathfrak{i}$ an ideal and $\mathfrak{s}$ a subalgebra. Then $\mathfrak{g}$ is the *semidirect sum* $\mathfrak{i} \rtimes \mathfrak{s}$, and $\mathfrak{s}$ acts on $\mathfrak{i}$ by the bracket.

Proposition 12.13 (The degree-two algebra)**.** *The polynomials of degree at most 2 in (q, p) are a Lie subalgebra of $C^\infty(\mathbb{R}^{2n})$ under $\{\cdot, \cdot\}$, of dimension $\frac{1}{2}(2n + 1)(2n + 2)$. It is the semidirect sum*

$$\mathfrak{h}_n \rtimes \mathfrak{sp}(2n), \tag{12.5}$$

with the Heisenberg algebra $\mathfrak{h}_n = \operatorname{span}\{1, q_i, p_i\}$ an ideal and the homogeneous quadratics $\cong \mathfrak{sp}(2n)$ acting on it. It is a direct sum of vector spaces, but the two summands do not commute under the bracket: $\{q^2/2, p\} = q$.

Proof. Proposition 5.17: a bracket of polynomials of degrees a and b has degree at most $a + b - 2$, so degree ≤ 2 is closed and the bracket of a quadratic with a linear function is linear, which makes $\mathfrak{h}_n$ an ideal. As a vector space, the polynomials of degree at most 2 split into $\mathfrak{h}_n$ and the homogeneous quadratics. The quadratics are $\mathfrak{sp}(2n)$ by Theorem 6.17. The dimension is the number of monomials of degree ≤ 2 in $2n$ variables. For the last clause, $\{q^2/2, p\} = q \neq 0$. ∎

On that subalgebra the dictionary is exact. Weyl's rule assigns to each polynomial of degree ≤ 2 a symmetrically ordered operator — $q^2 \mapsto Q^2$, $qp \mapsto \frac{1}{2}(QP+PQ)$ — and for two such polynomials

$$\widehat{\{f,g\}} = \frac{1}{\mathrm{i}\hbar}\left[\hat{f},\hat{g}\right] \qquad \text{exactly, with no correction in } \hbar. \tag{12.6}$$

The operators are built, and (12.6) is verified on a basis, in Berndt §§5.2–5.3 (pp. 116–125); we state the result and do not construct them.

The oscillator illustrates the correspondence. Take $n=1$ and $H=\frac{1}{2}(p^2+\nu^2q^2)$. Classically $\dot q=\{q,H\}=p$ and $\dot p=\{p,H\}=-\nu^2q$. With $[Q,P]=\mathrm{i}\hbar$ one has $\left[Q,P^2\right]=[Q,P]\,P+P\,[Q,P]=2\mathrm{i}\hbar P$ and likewise $\left[P,Q^2\right]=-2\mathrm{i}\hbar Q$, so the Heisenberg equations are

$$\dot Q=\frac{1}{\mathrm{i}\hbar}[Q,H]=P, \qquad \dot P=\frac{1}{\mathrm{i}\hbar}[P,H]=-\nu^2Q,$$

the same linear equations. The same formulas solve both: $Q_t=Q\cos\nu t+(P/\nu)\sin\nu t$ and $P_t=P\cos\nu t-\nu Q\sin\nu t$, with operators in place of numbers. Every bracket used here is between polynomials of degree at most 2.

Where the match stops

> **Theorem 12.14** (Groenewold–van Hove · *cited*)**.** *In one degree of freedom there is no linear map A from the polynomials in (q,p) to an associative algebra of operators with $A(1)=\mathbb{1}$, $A(q)=Q$ and $A(p)=P$ the Schrödinger operators, $A(q^2)=Q^2$, $A(p^2)=P^2$, and $A(\{f,g\})=\frac{1}{\mathrm{i}\hbar}[A(f),A(g)]$ for all polynomials f,g.*

Weyl's rule has every one of these properties on the polynomials of degree at most 2. So the exact correspondence (12.6) holds on the polynomials of degree at most 2, where the Hamiltonians of the oscillator and the free particle live, and in one degree of freedom it cannot be extended to all polynomials.

Summary.

- *Handed on to Integrability:* integrals in involution, at most n of them independent at a point (Definition 12.7 and Proposition 12.6); for n functions *in involution*, a level set at each point of which the $\mathrm{d}F_i$ are independent is Lagrangian and invariant under all n flows (Theorem 12.9); a compact connected component of it is a torus with actions $I_k=\frac{1}{2\pi}\oint_{\gamma_k}p\cdot\mathrm{d}q$ (Theorem 12.10, cited, and proved in Arnold).
- *The third answer:* the three layers side by side (§ 12.5). The Lagrange layer passes to the Hamilton layer for a hyperregular Lagrangian, and the Hamilton layer is a case of the Poisson layer, whose definition of a system is Definition 7.11.
- *Handed on to quantum mechanics:* the algebra, the states, and observables that generate automorphisms, for which Jacobi is the condition ((12.4), Theorem 6.5); not the points, since $[Q,P]=\mathrm{i}\hbar\,\mathbb{1}$ leaves no character; the exact correspondence on $\mathfrak{h}_n\rtimes\mathfrak{sp}(2n)$ (Proposition 12.13, (12.6)), which in one degree of freedom cannot be extended to all polynomials (Theorem 12.14).

References

Arnold, *Mathematical Methods of Classical Mechanics*, §§49–50 (pp. 271–285): Liouville's theorem in four parts (p. 272); the level set is null, item 1″ on p. 274 (our Theorem 12.9);

the actions (5) on p. 282, and their independence of the representative of the cycle on p. 283 (Figure 218). da Silva, *Lectures on Symplectic Geometry*, §18.4 (pp. 110–111): the remark before Definition 18.10 is Proposition 12.6. Berndt, *An Introduction to Symplectic Geometry*, §§5.2–5.3 (pp. 116–125): the operators of degree ≤ 2 and (12.6); Theorem 5.11 (p. 126): Groenewold–van Hove, with $A(q^2) = Q^2$, $A(p^2) = P^2$ as his hypothesis (13).